PROJECT MISSION MANUAL: AUTONOMOUS CROP ANOMALY INTERCEPTION ENGINE

MISSION: RS-SDA | ORBIT: SENTINEL-2 L2A | TARGET: ODISHA REGION
CLASSIFICATION: CRITICAL ENTERPRISE TECH

This technical manual serves as the comprehensive engineering handbook and system architecture layout for the repository deployed on GitHub. It outlines the end-to-end framework designed to ingest, filter, model, and autonomously intercept extreme vegetative deviations across target geographic nodes.

1. Executive Space-Mission Briefing

In satellite observation setups, monitoring precision depends heavily on how cleanly data flows through telemetry processing loops. This project establishes an autonomous computational asset that treats Earth Observation (EO) vegetation time-series as multi-frequency continuous telemetry feeds. By monitoring temporal vegetation indices over dense crop regions, the engine sets up an active control loop that identifies sudden localized structural degradation without human observation dependencies.

2. System Engineering & Pipeline Architecture

The operational framework transitions through four strict system phases inside the `MISSION_RS_SDA` space:

Phase A: Telemetry Harmonization & De-noising

Raw observations fetched from Google Earth Engine (GEE) contain systematic orbital and atmospheric variations. Temporal data gaps are resolved through deterministic linear resampling, while high-frequency scattering (caused by cloud edges and water vapor) is resolved utilizing a localized convolutional filter:

$$Y_j = \Sigma_{i=-m}^{m} C_i \cdot y_{j+i}$$

This implementation ensures that the core phenological velocity vectors remain structurally intact while stripping sensor irregularities.

Phase B: Sliding-Window State Discretization

To provide sufficient context for deep feature extraction, continuous historical data streams are formatted into high-dimensional matrix blocks. The system uses a fixed **30-Day Lookback Array (T_{-30} to T_0)** tracking historical growth velocity parameters, maps them against a **5-Day Prediction Horizon (T_{+1} to T_{+5})**, yielding optimized tensor dimensions of `(264, 30, 1)` for neural parsing.

Phase C: Stacked Neural Network Topology (LSTM)

Temporal dynamic sequences are captured using a Recurrent Deep Learning Architecture configured with 29,477 optimized trainable parameters. Feature extraction maps through dual-stacked layers:

Layer Type	Output Dimensionality	Regularization Matrix / Parameters
LSTM Layer 01 (Primary)	(None, 30, 64)	Stateful sequence returning vector configurations
Stochastic Dropout 01	(None, 30, 64)	20% Rate injection for atmospheric noise prevention
LSTM Layer 02 (Secondary)	(None, 32)	Deep temporal vector compression
Stochastic Dropout 02	(None, 32)	10% Rate injection ensuring generalization stability
Dense Projection Layer	(None, 5)	Linear transformation to 5-dimensional continuous prediction coordinates

SYSTEM CONVERGENCE METRICS

The model reached optimal system convergence across a 25-epoch sequence, logging a stabilized **Training MSE Loss of 0.0037** and a **Validation Loss of 0.0049**, proving there is zero overfitting within operational boundaries.

3. Statistical Interception Engine (2σ Threshold Logic)

The core automation relies on continuous operational inversion checking. The tracking loop calculates daily residual values between expected model profiles and raw observations:

$$Residual\ (\Delta) = Baseline_{Predicted} - Observation_{Actual}$$

When environmental anomalies trigger severe biomass drops, the variance escapes the historical standard deviation bounds:

$$\Delta > 2 \cdot \sigma \quad (where\ \sigma = 0.04,\ Threshold\ Barrier = 0.08)$$

Upon breaking this statistical limit, the system instantly stops normal logging loops, tracks the exact timestamp coordinate, and flags the anomaly state.

4. Enterprise-Grade Downstream Alert Serialization

Once a breach is intercepted, the edge engine dumps a machine-readable, production-standard tracking payload to the automation queue. Below is the telemetry asset structure exported by the script:


```
{
    "alert_id": "ALERT_ODISHA_PADDY_2026_09",
    "timestamp_generated": "2026-06-17 19:40:22",
    "target_field_id": "Field_1",
```

```
    "anomaly_type": "Severe Vegetative Stress / Potential Submergence",
    "severity_score": 0.4952,
    "historical_baseline_predicted_ndvi": 0.5452,
    "actual_satellite_observed_ndvi": 0.05,
    "system_escalation_status": "CRITICAL_ACTION_REQUIRED"
}
```


5. Repository Deployment Mapping

To reconstruct or plug into this mission architecture locally, the repository enforces a strict, isolated path hierarchy:

```
MISSION_RS_SDA/
└── Crop_Health_Monitor_Odisha/
    ├── notebooks/
    │   ├── 01_Data_Acquisition_GEE.ipynb
    │   ├── 02_Signal_Processing_Filtering.ipynb
    │   └── 03_LSTM_Anomaly_Interception_Engine.ipynb
    ├── models/
    │   ├── lstm_crop_anomaly_engine.keras       <-- Trained Neural Matrix Weights
    │   └── robust_minmax_scaler.pkl             <-- Spatial Scaling Reference Matrix
    └── outputs/
        └── alerts/
            └── automated_stress_alert.json      <-- Direct Telemetry Automated Payload
```